PROBLEM 31

# Two Particles on Cone with String

Original statement · Problems.pdf, p. 19

Multi-Particle Cable Constraint

Two heavy point particles $P_1$ and $P_2$ of mass $m$ each are connected by an inextensible, massless string of total length $\ell$. Particle $P_1$ is constrained to move without friction over the surface of a cone of semiangle $\alpha$ (the particle cannot leave the surface of the cone). The string passes through an aperture at the cone vertex $O$, and it can slide through it without friction. Particle $P_2$ hangs along the cone axis, and is constrained to move along it. We define an inertial reference frame $S_0$ as shown in the figure. We call $\xi$ the length of the fraction of string between $O$ and $P_1$, and $\theta$ the angle formed between the $Ox$ axis and the projection onto the $Oxy$ plane of the $\mathbf{OP}_1$ vector. In the following, assume that the string is always tense and that the particles do not reach point $O$ during motion.

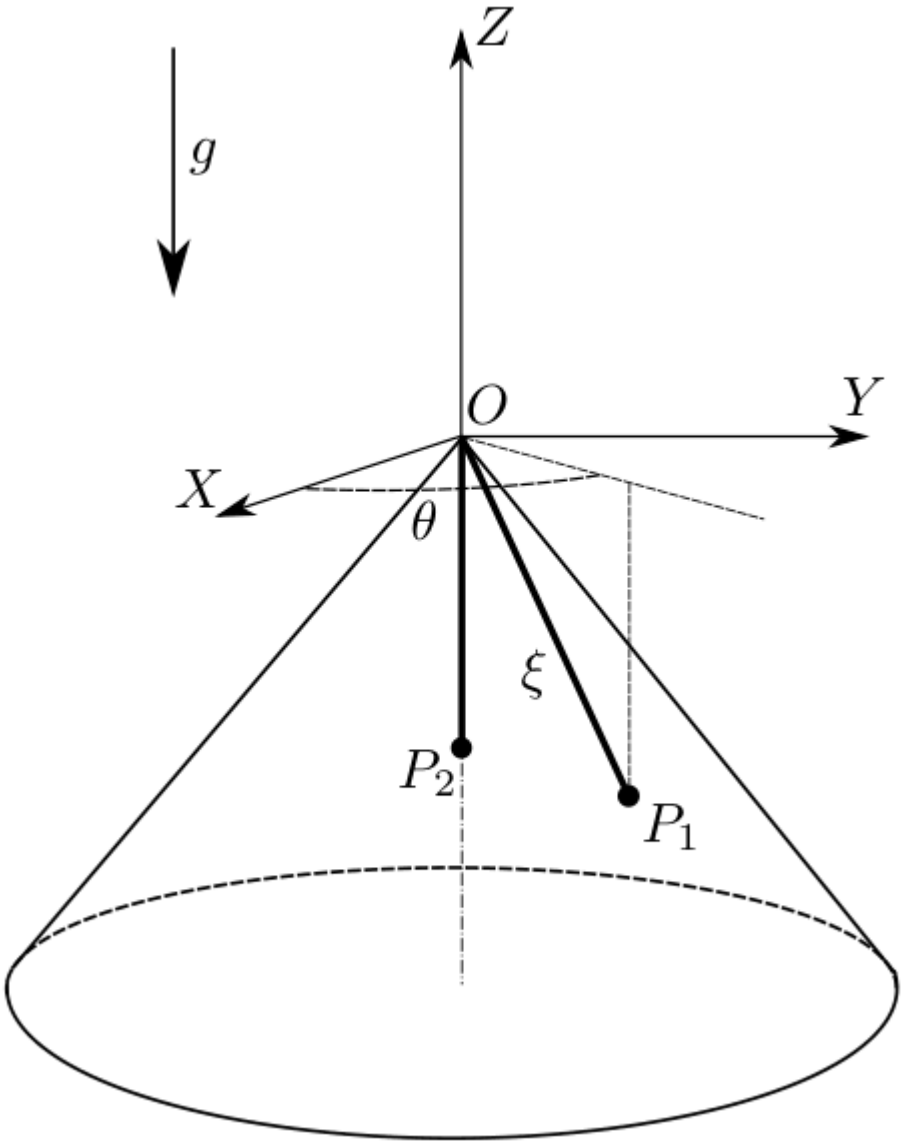



*Figure: Cone with vertex O and axes X, Y, Z; P1 on the cone surface at string length ξ and angle θ, P2 hanging along the axis (Problems.pdf, p. 19).*

**(a)** Write down the general expression of the position, velocity and acceleration vectors of $P_1$ and $P_2$ as a function of $\theta, \xi$ and the parameters of the problem.

**(b)** Write down the expression of all forces that act on $P_1$ and $P_2$.

**(c)** Write down the differential equations that, if solved, would yield the time evolution of $\theta, \xi$, the tension on the string, and any other reaction force in the problem.

**(d)** What is the mathematical condition that the function $\xi(t)$ must satisfy for the string to remain tense all the time?

**(e)** How fast should $P_1$ rotate in the $\theta$ direction to have a circular trajectory over a cone parallel? What is the value of the string tension and any other reaction force in the problem in that case? Give your answer as a function of $\xi$ and the parameters of the problem.

**(f)** Taking these equations as the starting point (or based on the physical considerations of the problem) write down two first-integrals of the problem that do not depend on the string tension or any other reaction force in the problem.

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**Worked Solution (4-Phase Methodology)**

PHASE 1

### Constraints, Degrees of Freedom & Coordinates

- **System:** two heavy particles $P_1$ and $P_2$ of mass $m$ are joined by an inextensible massless string of length $\ell$ that passes without friction through the aperture at the vertex $O$ of a cone of semi-angle $\alpha$. $P_1$ moves without friction on the cone surface; $P_2$ hangs along the cone axis ($Oz$ vertical upward, the cone opens downward). The string stays taut and the particles do not reach $O$.
- **Constraints:** with 6 coordinates for two free particles:
  - $P_1$ on the cone: 1 constraint (bilateral: the statement says the particle cannot leave the surface).
  - Constant string length: $\xi + (\text{depth of } P_2) = \ell$, i.e. 1 constraint (unilateral: $T \geq 0$).
  - $P_2$ on the axis: $x_2 = y_2 = 0$, 2 constraints.
- **Degrees of freedom:** $\text{CDOF} = 6 - 4 = 2$, with $\xi$ (the length of the string between $O$ and $P_1$) and $\theta$ (the azimuthal angle of $P_1$).
- **Parameters:** $m, \ell, \alpha, g$.

PHASE 2

### Reference Frames & Vector Bases

- **Inertial frame:** $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ with $\mathbf{k}$ vertical upward.
- **Cylindrical basis:** $\mathbf{e}_R = \cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}$, $\mathbf{e}_\theta = -\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}$, $\mathbf{k}$ (rotation about $\mathbf{k}$ by $\theta$ ; $\det = 1$, $RR^T = I$; $\dot{\mathbf{e}}_R = \dot\theta\,\mathbf{e}_\theta$, $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_R$).
- **Cone basis:** the unit vector along the generator from $O$ to $P_1$, $\mathbf{e}_\xi = \sin\alpha\,\mathbf{e}_R - \cos\alpha\,\mathbf{k}$; the unit outward normal to the cone, $\mathbf{n} = \cos\alpha\,\mathbf{e}_R + \sin\alpha\,\mathbf{k}$; and $\mathbf{e}_\theta$. The triple $\{\mathbf{e}_\xi, \mathbf{e}_\theta, \mathbf{n}\}$ is right-handed ($\mathbf{e}_\xi \times \mathbf{e}_\theta = \mathbf{n}$).

PHASE 3

## Derivation

### 3.1 Kinematics

#### (a) Position, velocity and acceleration

**Pedagogical Justification:** $P_1$ is at the distance $\xi$ from $O$ along the generator, so $\mathbf{r}_1 = \xi\,\mathbf{e}_\xi = \xi\sin\alpha\,\mathbf{e}_R - \xi\cos\alpha\,\mathbf{k}$. Since $\mathbf{e}_R$ rotates with $\theta$, the derivatives use the product rule and Poisson's formula. $P_2$ is on the axis at the depth $\ell - \xi$ below $O$ (the string has length $\xi$ on the cone side and $\ell - \xi$ on the hanging side).

$$\mathbf{v}_1 = \dot\xi\,\mathbf{e}_\xi + \xi\sin\alpha\,\dot\theta\,\mathbf{e}_\theta$$

$$\mathbf{a}_1 = \left(\ddot\xi - \xi\dot\theta^2\sin^2\alpha\right)\mathbf{e}_\xi + \sin\alpha\left(\xi\ddot\theta + 2\dot\xi\dot\theta\right)\mathbf{e}_\theta - \xi\dot\theta^2\sin\alpha\cos\alpha\,\mathbf{n}$$

$$\mathbf{r}_2 = -(\ell - \xi)\,\mathbf{k}, \qquad \mathbf{v}_2 = \dot\xi\,\mathbf{k}, \qquad \mathbf{a}_2 = \ddot\xi\,\mathbf{k}$$

### 3.2 Forces

**(b) Forces on $P_1$ and $P_2$**

- **On $P_1$:** the weight $-mg\,\mathbf{k} = mg\cos\alpha\,\mathbf{e}_\xi - mg\sin\alpha\,\mathbf{n}$; the tension of the string, directed toward $O$, $\mathbf{T}_1 = -T\,\mathbf{e}_\xi$ ($T > 0$); and the reaction of the cone $\mathbf{N} = N\,\mathbf{n}$ (outward).
- **On $P_2$:** the weight $-mg\,\mathbf{k}$; the tension $+T\,\mathbf{k}$ (same string, massless, smooth aperture); and the reaction $\mathbf{R}_2$ of the guide, perpendicular to $\mathbf{k}$ (horizontal).
- **Inertia forces:** none ($S_0$ is inertial).

### 3.3 Equations of Motion

**(c) Differential equations**

> **Pedagogical Justification:** Newton's second law for each particle, six scalar equations for the six unknowns $\xi, \theta, T, N$ and the two horizontal components of $\mathbf{R}_2$:

$$P_2,\ \mathbf{k}:\ m\ddot{\xi} = T - mg \implies T = m\left(g + \ddot{\xi}\right), \qquad P_2,\ \text{horizontal}:\ \mathbf{R}_2 = \mathbf{0}\ (\text{since } \mathbf{a}_2 \parallel \mathbf{k})$$

$$P_1,\ \mathbf{e}_\xi:\ m\left(\ddot{\xi} - \xi\dot{\theta}^2\sin^2\alpha\right) = mg\cos\alpha - T$$

$$P_1,\ \mathbf{e}_\theta:\ m\sin\alpha\left(\xi\ddot{\theta} + 2\dot{\xi}\dot{\theta}\right) = 0, \qquad P_1,\ \mathbf{n}:\ -m\xi\dot{\theta}^2\sin\alpha\cos\alpha = N - mg\sin\alpha$$

- **Decoupling.** Substituting $T = m(g + \ddot{\xi})$ in the $\mathbf{e}_\xi$ equation eliminates the tension and leaves the equations of the two free coordinates:

$$\boxed{2\ddot{\xi} - \xi\dot{\theta}^2\sin^2\alpha + g\,(1 - \cos\alpha) = 0}, \qquad \boxed{\xi\ddot{\theta} + 2\dot{\xi}\dot{\theta} = 0}$$

and, once $\xi(t)$ and $\theta(t)$ are known,

$$T = m\left(g + \ddot{\xi}\right), \qquad N = mg\sin\alpha - m\xi\dot{\theta}^2\sin\alpha\cos\alpha$$

- **First integrals** (independent of $T$ and $N$): see 3.4(f).

**3.4 Integration and Answers**

**(d) Condition for the string to remain taut**

**Pedagogical Justification:** the string can only pull, so $T > 0$ at all times:

$$T = m\left(g + \ddot{\xi}\right) > 0 \iff \boxed{\ddot{\xi} > -g}$$

Substituting $\ddot{\xi} = \frac{1}{2}\left[\xi\dot{\theta}^2\sin^2\alpha - g(1-\cos\alpha)\right]$ from the decoupled equation of (c) gives $T = \frac{m}{2}\left[g\left(1+\cos\alpha\right) + \xi\dot{\theta}^2\sin^2\alpha\right] > 0$ for $0 < \alpha < \pi$, so the string stays taut throughout the motion.

**Discrepancy with the official solution**
The key of `Problems.pdf` states the condition as $\ddot{\xi} > g$. That cannot be right: for the horizontal circular trajectory of part (e) the key itself gives $T = mg > 0$ with $\ddot{\xi} = 0$, which violates $\ddot{\xi} > g$, while $\ddot{\xi} = 0 > -g$ is satisfied. The equation of $P_2$ gives $T = m(g + \ddot{\xi})$, hence $\ddot{\xi} > -g$ (the minus sign is missing in the key).

**(e) Circular trajectory on a parallel**

**Pedagogical Justification:** a horizontal circle on the cone has $\xi$ constant, so $\dot{\xi} = \ddot{\xi} = 0$ and the radial equation reduces to

$$-\xi\dot{\theta}^2\sin^2\alpha + g\left(1-\cos\alpha\right) = 0 \implies \boxed{\dot{\theta} = \frac{1}{\sin\alpha}\sqrt{\frac{g\left(1-\cos\alpha\right)}{\xi}}}$$

- **Tension:** $T = m\,(g + 0) = mg$, directed on $P_1$ as $\mathbf{T}_1 = -mg\,\mathbf{e}_\xi = mg\,(-\sin\alpha\,\mathbf{e}_R + \cos\alpha\,\mathbf{k})$.
- **Cone reaction:** with $\xi\dot\theta^2\sin^2\alpha = g\,(1-\cos\alpha)$:

$$N = mg\sin\alpha - m\cos\alpha\,\frac{g\,(1-\cos\alpha)}{\sin\alpha} = \frac{mg}{\sin\alpha}\left(\sin^2\alpha - \cos\alpha + \cos^2\alpha\right) = \boxed{\frac{mg\,(1-\cos\alpha)}{\sin\alpha}}, \qquad \mathbf{N} = N\,(\cos\alpha\,\mathbf{e}_R + \sin\alpha\,\mathbf{k})$$

$N > 0$ for $0 < \alpha < \pi$: the cone presses on the particle (which would also be admissible for a one-sided contact).

**(f) Two first integrals independent of the tension and the reactions**

- **Azimuthal integral (first integral of the $\mathbf{e}_\theta$ equation).** With the product rule, the azimuthal acceleration is $\sin\alpha\left(\xi\ddot\theta + 2\dot\xi\dot\theta\right) = \dfrac{\sin\alpha}{\xi}\dfrac{d}{dt}\left(\xi^2\dot\theta\right)$. Gravity, the tension (toward $O$) and the cone reaction all lie in the meridian plane of $P_1$, so they have no $\mathbf{e}_\theta$ component:

$$\frac{d}{dt}\left(\xi^2\dot\theta\right) = 0 \implies \boxed{\xi^2\dot\theta = C_1}$$

- **Energy.** The reactions do no work: $\mathbf{N}\cdot\mathbf{v}_1 = 0$ ($\mathbf{N}$ is normal to the surface) and the tension does the work $-T\dot\xi$ on $P_1$ and $+T\dot\xi$ on $P_2$, which cancel (inextensible string, smooth aperture). Gravity is conservative. The kinetic and potential energies are

$$K = \tfrac{1}{2}m\left(\dot\xi^2 + \xi^2\dot\theta^2\sin^2\alpha\right) + \tfrac{1}{2}m\dot\xi^2 = m\dot\xi^2 + \tfrac{1}{2}m\xi^2\dot\theta^2\sin^2\alpha, \qquad V = -mg\xi\cos\alpha - mg(\ell - \xi) = mg\xi\,(1-\cos\alpha) - mg\ell$$

$$\boxed{2\dot\xi^2 + \xi^2\dot\theta^2\sin^2\alpha + 2g\xi\,(1-\cos\alpha) = C_2}$$

where the constant $C_2$ absorbs $-2g\ell$ and the factor $2/m$.

PHASE 4

### Interpretation & Checks

- **Effective potential.** With $\xi^2\dot\theta = C_1$ the energy becomes $2\dot\xi^2 + W(\xi) = C_2$ with $W(\xi) = \dfrac{C_1^2 \sin^2\alpha}{\xi^2} + 2g\xi\,(1-\cos\alpha)$. Its minimum is at $W'(\xi_c) = -\dfrac{2C_1^2\sin^2\alpha}{\xi_c^3} + 2g\,(1-\cos\alpha) = 0$, i.e. $\xi_c\dot\theta^2\sin^2\alpha = g\,(1-\cos\alpha)$, the circular parallel of (e); and $W'' = \dfrac{6C_1^2\sin^2\alpha}{\xi^4} > 0$, so it is a stable minimum.
- **Dimensional check:** $[\xi^2\dot\theta] = \mathrm{m^2 s^{-1}}$ and $[g\xi] = \mathrm{m^2 s^{-2}} = [\dot\xi^2]$; $[T] = [N] = \mathrm{N}$.
- **Check against the official key:** the key of `Problems.pdf` gives two degrees of freedom, the quadratures $\xi^2\dot\theta = C_1$ and $2\dot\xi^2 + \xi^2\dot\theta^2\sin^2\alpha + 2g\xi(1-\cos\alpha) = C_2$, the circular condition $\xi\dot\theta^2\sin^2\alpha = g(1-\cos\alpha)$, $\mathbf{T} = mg\,(-\sin\alpha\,\mathbf{e}_R + \cos\alpha\,\mathbf{k})$ and $\mathbf{N} = \dfrac{mg(1-\cos\alpha)}{\sin\alpha}(\cdots)$. Everything agrees with the results above, with two remarks: the tension condition of the key ($\ddot\xi > g$) is an erratum (see (d)), and the key writes the direction of $\mathbf{N}$ with $\mathbf{e}_\theta$, which is tangent to the parallel and cannot be normal to the cone; the normal direction is $\cos\alpha\,\mathbf{e}_R + \sin\alpha\,\mathbf{k}$. Sympy confirms the three components of $\mathbf{a}_1$ and numerical integration of the equations of (c) ($\alpha = 40^\circ$) conserves $\xi^2\dot\theta$ and the energy to $3\times10^{-14}$, with $T$ and $N$ positive throughout and $N/(mg) = 0.363970$ on the circular orbit, equal to $(1-\cos\alpha)/\sin\alpha$.

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*Source: Problems.pdf*